

D441

Medical Dosage Calculations and Pharmacology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 6-year-old, whose family just moved from another state and who is allergy-free, has an order for amoxicillin 400 mg by mouth. The pharmacy sends a suspension labeled 250 mg per 5 mL. How many mL should the nurse give?

- A. 1.6 mL
- B. 8 mL
- C. 5 mL
- D. 3.1 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A nurse gives 3 tablets to deliver an ordered 150 mg dose. What is the strength of each tablet?

- A. 450 mg
- B. It cannot be found without the patient's weight.
- C. 50 mg
- D. 150 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A child who plays on a soccer team and weighs 44 lb has an order for a drug at 10 mg/kg/day, divided every 8 hours. The child's mother is a pharmacist. How many mg should the nurse give per dose?

- A. 146.7 mg
- B. 600 mg
- C. 66.7 mg
- D. 200 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A 44-lb child has an order for 100 mg every 6 hours. The safe maximum is 15 mg/kg/day. Is the order within the safe range?

- A.** Yes; the maximum is 660 mg/day.
- B.** Yes; 100 mg is below the 300 mg maximum.
- C.** No; any pediatric dose over 100 mg is unsafe.
- D.** No; 400 mg/day exceeds the 300 mg/day maximum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A heparin bag contains 25,000 units in 250 mL. The order is 1,200 units per hour. The patient, a long-haul truck driver, was admitted at 0300 with a DVT. At what rate should the pump be set?

- A.** 1,200 mL/hr
- B.** 4.8 mL/hr
- C.** 12 mL/hr
- D.** 0.08 mL/hr

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A pump is running a heparin bag of 25,000 units in 250 mL at 15 mL/hr. How many units per hour is the patient receiving?

- A.** 0.15 units/hr
- B.** 15 units/hr
- C.** 1,500 units/day
- D.** 1,500 units/hr

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A patient, who is a retired pilot and walks two miles a day, receives a single 200 mg dose of a drug with a half-life of 6 hours. Kidney function is normal. How much drug remains after 18 hours?

- A. 25 mg
- B. 66.7 mg
- C. 50 mg
- D. 0 mg; the drug is fully gone after three half-lives.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A drug level falls from 80 mg/L to 10 mg/L in 12 hours. What is the half-life?

- A. 4 hours
- B. 12 hours
- C. 1.5 hours
- D. 6 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

At 1400 on a busy surgical unit, a nurse draws up 2 mg of morphine from a 4 mg vial for a patient who rates pain 6/10. The charge nurse is in a meeting. What must happen to the remaining 2 mg?

- A. Waste it alone and document the amount.
- B. Save it until the charge nurse returns to co-sign.
- C. Have a nursing assistant witness and sign.
- D. Another licensed nurse watches the waste and co-signs right then.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A patient on IV vancomycin every 12 hours, who is a high-school coach, has a trough level ordered. The next dose is due at 2100. The lab opens a new analyzer that week. When should the level be drawn?

- A.** Within about 30 minutes before the 2100 dose
- B.** Two hours after the morning dose
- C.** Any time, since the level is steady on IV therapy
- D.** Right after the 2100 infusion finishes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A patient with hypertension, whose blood pressure is 128/76 and who watches baseball every evening, has metoprolol due at 0900. The nurse counts an apical pulse of 52. The order says hold if HR under 60. What should the nurse do?

- A.** Give half the dose.
- B.** Hold the dose and notify the provider.
- C.** Give the dose and recheck the pulse in an hour.
- D.** Give the dose, because the blood pressure is normal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A 70-year-old retired librarian taking warfarin, who lives alone with two cats, tells the nurse she started ginkgo biloba to improve her memory. What is the main concern?

- A.** Reduced warfarin effect and higher clot risk
- B.** Increased bleeding risk
- C.** None, because herbal products are natural
- D.** Serotonin syndrome

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A 6-year-old, ~~whose family just moved from another state and who is allergy free~~, has an order for amoxicillin 400 mg by mouth. The pharmacy sends a suspension labeled 250 mg per 5 mL. How many mL should the nurse give?

Target word: volume to give **Decides:** $(400 \div 250) \times 5$ mL.

A	Dropped element	Stops at the ratio and drops the 5 mL volume.
B	Correct	$400 \div 250 = 1.6$; $1.6 \times 5 \text{ mL} = 8 \text{ mL}$.
C	Wrong stem number	The label volume, not the dose volume.
D	Inverted operation	Divides on-hand by desired: $250 \div 400 \times 5$.

2. Answer C Concept 1 • transfer

A nurse gives 3 tablets to deliver an ordered 150 mg dose. What is the strength of each tablet?

Target word: solve for the on-hand strength **Decides:** $150 \text{ mg} \div 3$ tablets.

A	Inverted operation	Multiplies instead of dividing.
B	Adjacent category	Weight matters for mg/kg orders, not for this fixed dose.
C	Correct	$150 \div 3 = 50 \text{ mg per tablet}$.
D	Wrong stem number	The total dose, not the per-tablet strength.

3. Answer C Concept 2

A child ~~who plays on a soccer team~~ and weighs 44 lb has an order for a drug at 10 mg/kg/day, divided every 8 hours. ~~The child's mother is a pharmacist.~~ How many mg should the nurse give per dose?

Target word: mg per dose **Decides:** $44 \text{ lb} \div 2.2 = 20 \text{ kg}$; $10 \times 20 = 200 \text{ mg/day}$; $\div 3$ doses.

A	Confusion set	Uses pounds: $44 \times 10 = 440 \div 3$.
B	Inverted operation	Multiplies the daily dose by 3 instead of dividing.
C	Correct	$20 \text{ kg} \times 10 = 200 \text{ mg/day} \div 3 \text{ doses} \approx 66.7 \text{ mg}$.
D	Dropped element	The daily total; drops the division into three doses.

4. Answer D Concept 2 • transfer

A 44-lb child has an order for 100 mg every 6 hours. The safe maximum is 15 mg/kg/day. Is the order within the safe range?

Target word: check daily total against daily max **Decides:** $20 \text{ kg} \times 15 = 300 \text{ mg/day max}$; order = 400 mg/day.

A	Confusion set	Uses pounds: 44×15 .
B	Dropped element	Compares one dose with a daily limit.
C	Absolute language	Safety depends on mg/kg, not a fixed cutoff.
D	Correct	$100 \text{ mg} \times 4 \text{ doses} = 400 \text{ mg/day} > 300$.

5. Answer C Concept 3

A heparin bag contains 25,000 units in 250 mL. The order is 1,200 units per hour. The patient, ~~a long-haul truck driver, was admitted at 0300~~ with a DVT. At what rate should the pump be set?

Target word: pump rate mL/hr **Decides:** Concentration = $25,000 \div 250 = 100$ units/mL.

A	Wrong stem number	The dose figure used as a rate.
B	Confusion set	Divides by the bag volume ($1,200 \div 250$) instead of the concentration.
C	Correct	$1,200 \text{ units/hr} \div 100 \text{ units/mL} = 12 \text{ mL/hr}$.
D	Inverted operation	Divides concentration by dose.

6. Answer D Concept 3 • transfer

A pump is running a heparin bag of 25,000 units in 250 mL at 15 mL/hr. How many units per hour is the patient receiving?

Target word: solve for dose from rate **Decides:** $100 \text{ units/mL} \times 15 \text{ mL/hr}$.

A	Inverted operation	Divides the rate by the concentration.
B	Wrong stem number	The pump rate, not the dose.
C	Adjacent category	Right number, wrong time unit.
D	Correct	$100 \text{ units/mL} \times 15 \text{ mL/hr} = 1,500$.

7. Answer A Concept 4

A patient, ~~who is a retired pilot and walks two miles a day,~~ receives a single 200 mg dose of a drug with a half-life of 6 hours. Kidney function is normal. How much drug remains after 18 hours?

Target word: amount remaining **Decides:** $18 \text{ h} = 3 \text{ half-lives}$: $200 \rightarrow 100 \rightarrow 50 \rightarrow 25$.

A	Correct	Three halvings of 200 mg leave 25 mg.
B	Inverted operation	Divides by the number of half-lives instead of halving three times.
C	Dropped element	Stops after two half-lives.
D	Absolute language	About 4–5 half-lives remove most of a drug, and never all of it at once.

8. Answer A Concept 4 • transfer

A drug level falls from 80 mg/L to 10 mg/L in 12 hours. What is the half-life?

Target word: solve for half-life **Decides:** $80 \rightarrow 40 \rightarrow 20 \rightarrow 10 = 3 \text{ half-lives}$ in 12 h.

A	Correct	$12 \text{ h} \div 3 \text{ half-lives} = 4 \text{ h}$.
B	Wrong stem number	The total elapsed time.
C	Inverted operation	Divides 12 by the 8-fold drop instead of by the number of halvings.
D	Dropped element	Counts only two halvings.

9. Answer D Concept 5

At 1400 on a busy surgical unit, a nurse draws up 2 mg of morphine from a 4 mg vial for a patient ~~who rates pain 6/10. The charge nurse is in a meeting.~~ What must happen to the remaining 2 mg?

Target word: waste procedure **Decides:** Controlled-substance waste needs a licensed witness at the time.

A	Dropped element	Drops the required witness.
B	Camouflage	Uses the charge-nurse detail; any available licensed nurse can witness, and holding a drug unsecured is unsafe.
C	Adjacent category	The witness must be a licensed nurse (or other authorized licensed staff).
D	Correct	Real-time witnessed waste protects against diversion.

10. Answer A Concept 6

A patient on IV vancomycin every 12 hours, ~~who is a high school coach,~~ has a trough level ordered. The next dose is due at 2100. ~~The lab opens a new analyzer that week.~~ When should the level be drawn?

Target word: trough timing **Decides:** Trough = lowest level, just before the next dose.

A	Correct	The trough reflects the lowest level right before the next dose.
B	Adjacent category	That is neither a true peak nor trough timing.
C	Absolute language	Levels rise and fall between doses; timing matters.
D	Inverted operation	That timing measures a peak, not a trough.

11. Answer B Concept 7

A patient with hypertension, ~~whose blood pressure is 128/76 and who watches baseball every evening,~~ has metoprolol due at 0900. The nurse counts an apical pulse of 52. The order says hold if HR under 60. What should the nurse do?

Target word: act on hold parameter **Decides:** Apical pulse 52 is below the hold limit of 60.

A	Adjacent category	Changing the dose is outside the nurse's role.
B	Correct	HR below the ordered limit means hold and report.
C	Dropped element	Drops the hold parameter.
D	Camouflage	Leans on the normal blood pressure; the pulse decides.

12. Answer B Concept 8

A 70-year-old retired librarian taking warfarin, ~~who lives alone with two cats,~~ tells the nurse she started ginkgo biloba to improve her memory. What is the main concern?

Target word: herb-drug interaction **Decides:** Ginkgo adds antiplatelet effect to an anticoagulant.

A	Inverted operation	Reverses the direction of the interaction.
B	Correct	Ginkgo inhibits platelets and adds to warfarin's effect.
C	Absolute language	Natural products can have strong drug interactions.
D	Confusion set	That risk belongs to St. John's wort with serotonergic drugs.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Desired over have	Amount to give = (desired dose ÷ dose on hand) × quantity it comes in.	Dividing on-hand by desired, or forgetting the volume the dose is dissolved in.	An order in mg and a supply labeled mg per tablet or mg per mL.
2	Weight-based dosing	Convert pounds to kilograms (÷ 2.2), multiply by mg/kg, then split a daily dose by the number of doses; compare daily totals to daily limits.	Using pounds in a mg/kg formula or comparing one dose to a daily maximum.	mg/kg/day, “divided every 8 hours,” or a weight in pounds.
3	IV infusion rate from a dose	mL/hr = dose per hour ÷ concentration (units or mg per mL).	Dividing by the bag volume instead of the concentration.	Units/hr or mg/hr ordered with a bag of drug in a given volume.
4	Half-life	Each half-life cuts the amount in the body by half; after 4-5 half-lives the drug is nearly gone.	Subtracting a fixed amount each interval or dividing by the number of half-lives.	A half-life in hours and an elapsed time.
5	Controlled substance accountability	Wasting a controlled drug requires a second licensed nurse to witness and co-sign at the time of the waste.	Documenting a waste alone or having an assistant or late witness sign.	A partial dose of an opioid or benzodiazepine.
6	Peak and trough timing	A trough is drawn just before the next dose; a peak is drawn after the dose has been absorbed or infused.	Drawing a trough right after the dose.	Vancomycin or aminoglycoside levels and “when should the nurse draw?”
7	Hold parameters by drug class	Check the apical pulse before beta blockers and digoxin; hold and notify if below the ordered limit (often 60).	Giving a scheduled cardiac drug without checking or halving the dose on one’s own.	A heart rate near 60 with a cardiac medication due.
8	Complementary therapy interactions	Herbal products interact with drugs: ginkgo, garlic, and ginger raise bleeding risk with anticoagulants; St. John’s wort raises serotonin syndrome risk and lowers many drug levels.	Assuming “natural” products have no drug interactions.	A patient on a prescription who mentions an herbal supplement.

Course scope (public description): Medical Dosage Calculations and Pharmacology introduces an in-depth nursing approach to medication administration concepts, legal & ethical principles, pharmacological principles, variety of drug classifications, complementary & alternative therapies needed to care for diverse patients across the lifespan.